

Summary on elementary matrices

For each type of elementary matrix, below is an example together with its inverse.

a.) $\begin{pmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ inverse: $\begin{pmatrix} 1 & -5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

b.) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ inverse: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/7 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

c.) $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ inverse: $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$
self-inverse

Swaps rows 2 and 3 in 4×1 , 4×2 , 4×3 , 4×4 , 4×5 , 4×6 , etc. matrices.

These are the only types of elementary matrices.

They are always easy to recognize and easy to invert.

Recall:

If E is an elementary matrix corresponding to elementary operation e then

$$e(A) = E \cdot A$$

matrix obtained
after application
of e to A

In this order!

Definition Let A and B of the same size. Then A is row-equivalent to B if B can be obtained from A by a sequence of row operations, i.e., if there exist row operations $e_1, e_2, \dots, e_k$ s.t.

$$\xleftarrow{e_k(\dots e_3(e_2(e_1(A)))\dots)} = B.$$

Equivalently, if there exist elementary matrices $E_1, E_2, \dots, E_k$ s.t.

$$\xleftarrow{E_k \dots E_2 E_1} A = B.$$

Fact Row equivalence is an equivalence relation.

Ex. If $\xleftarrow{E_k \dots E_2 E_1} A = B$ then $\xrightarrow{E_1^{-1} E_2^{-1} \dots E_k^{-1}} B = A.$

Hence if A is row-equivalent to B then B is row-equivalent to A .

Theorem 1 Every matrix A is row equivalent to a matrix B is row-echelon form.

Proof. Gauss reduction $\square$

Theorem 2 If an $n \times n$ matrix A is row equivalent to I_n then A can be written as a product of elementary matrices.

Proof If A is row equivalent to I_n then there exist elementary matrices $E_1, E_2, \dots, E_k$ such that

$$\xleftarrow{E_k \dots E_2 E_1} A = I_n.$$

We bring the elementary matrices to the right-hand side (in opposite order; as inverses) and get

$$\begin{aligned} A &= E_1^{-1} E_2^{-1} \dots E_k^{-1} I_n \\ &= E_1^{-1} E_2^{-1} \dots E_k^{-1} \end{aligned}$$

$\square$

Theorem 3 Suppose U is an $n \times n$ matrix in row echelon form with $u_{ii} \neq 0$ for all i . Then U is row equivalent to the identity matrix.

Proof U must look like this:

$$U = \begin{pmatrix} \textcircled{u_{11}} & u_{12} & \dots & \dots & u_{1n} \\ 0 & \textcircled{u_{22}} & \dots & \dots & u_{2n} \\ 0 & 0 & \textcircled{u_{33}} & \dots & u_{3n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & \textcircled{u_{nn}} \end{pmatrix}$$

all non-zero (pivots)

1.) multiply each row i by $\frac{1}{u_{ii}}$

2.) Subtract a suitable multiple of row i from the rows above it to get 0s (also) above the diagonal.

We end up with I_n .

□

Theorem 4 Suppose A is an invertible $n \times n$ matrix.
Then for every $\mathbf{b} \in \mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution.

Proof This is easy:

Existence: Plug in $A^{-1}\mathbf{b}$ for $\mathbf{x}$ in the equation.

No more than one solution: Multiply both sides of the equation by A^{-1} on the left.

We get $A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{b}$

It follows that

$$\underline{\mathbf{x} = A^{-1}\mathbf{b}}$$

□

Remark: You have done this in high school with numbers instead of matrices.

You want to solve

$$5x = 7.$$

You multiply both sides of the equation with the multiplicative inverse of 5 and get

$$\frac{1}{5} \cdot 5x = \frac{1}{5} \cdot 7,$$

which implies

$$\underline{x = \frac{7}{5}.$$

Hence infin. many solutions,
and therefore free variables.

Definition A matrix A is singular if the equation $A\mathbf{x} = \mathbf{0}$ has more than one solution (not only the trivial solution $\mathbf{x} = \mathbf{0}$).

Ex $A = \begin{pmatrix} 1 & 2 \\ 0 & 5 \end{pmatrix}$ is non-singular:

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

implies

$$\underline{x_2 = 0} \text{ and } \underline{x_1 = 0}.$$

Ex. $\begin{pmatrix} x_1 & x_2 & x_3 \\ 1 & 0 & 4 \\ 2 & 0 & 5 \\ 3 & 0 & 6 \end{pmatrix}$ is singular because of the zero column.

We can take any value for x_2 . To see this, consider the matrix times vector multiplication as linear combination of the columns in A .

Ex. $\begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 2 \\ 3 & 1 & 3 \end{pmatrix}$ is singular (two equal columns)

Ex. $\begin{pmatrix} 2 & 1 & 1 \\ 4 & 0 & 2 \\ 6 & 1 & 3 \end{pmatrix}$ is singular (one column is twice the other).

In general, it is not so easy to see if a matrix is singular or not. You need to determine a row echelon form and check if every column has a pivot.

Theorem 5 If A and B are row equivalent $m \times n$ matrices then the equations $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have the same solution set.

Proof If A and B are row equivalent then there are elementary matrices $E_1, E_2, \dots, E_k$ such that $E_k \dots E_2 E_1 A = B$.

Suppose we have $A\mathbf{x} = \mathbf{0}$. It follows that $\underbrace{E_k \dots E_2 E_1 A}_{B} \mathbf{x} = \underbrace{E_k \dots E_2 E_1}_{\text{same}} \mathbf{0} = \mathbf{0}$

Hence we have $B\mathbf{x} = \mathbf{0}$.

The other direction can be shown analogously using the inverses of the elementary matrices. $\square$

Theorem 6 Let A be a non-singular $n \times n$ matrix. Then A is row-equivalent to I_n .

Proof Let A be non-singular and U any row-echelon form of A . It follows that all diagonal entries in U are non-zero. This implies that we can continue reducing U to the identity matrix I_n (Thm 3). $\square$